PKM robot for respiratory simulation

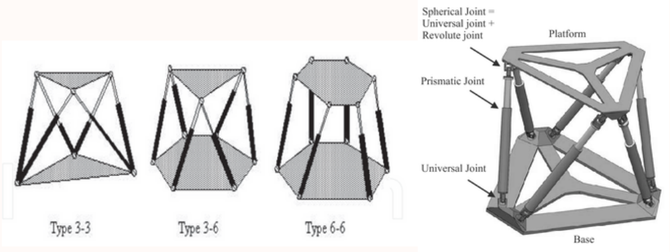
Shobit Verma

DESIGN PRESENTATION

Brief Background for PKM

- PKMs or parallel robots are the manipulators which having the end effector connected to the ground via several chains having actuations in parallel will have greater rigidity and superior positioning capability
- Used for applications where high load carrying capacity, good dynamic performance and precise positioning are of paramount importance, like flight simulator
- There are various configurations of this platform, but we would be looking at the 6 UPS (or 6 SPS) or commonly called the Stewart platform
- PKMs are particularly proposed for good ratios of strength-to-weight.
- the topology also suffers from its complex and small work-space, kinematic singularities and inconsistent dexterity which also limits its flexibility

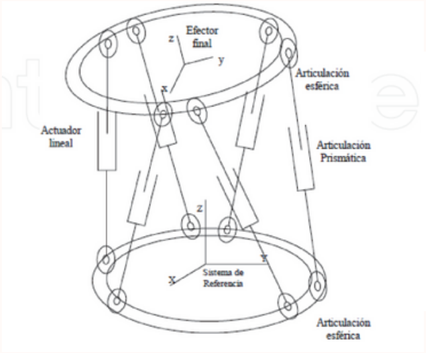
[1]



[2]

[3]

[4]



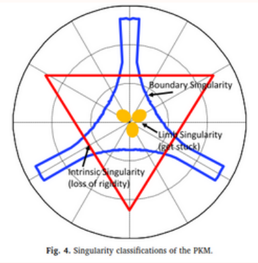


Fig. 4. Singularity classifications of the PKM.

REFERENCES

The Stewart platform manipulator: a review Bhaskar Dasguptaa,*, T.S. Mruthyunjaya [1]

Stewart-Gough Platform: Design and Construction with a Digital PID Controller Implementation Flabio Dario Mirelez-Delgado, José Ronaldo etc. [2]

Tlusty, J. Ziegert, S. Ridgeway, Fundamental comparison of the use of serial and parallel kinematics for machines tools, CIRP Ann. Manuf. Technol. 48 [3]

Optimization design for a compact redundant hybrid parallel kinematic machine Yen Ping-Lang a, Wang Ching-Henga, Lin Hao-Tingb, Hung Shuo-Sueic,d [4]

Existing works on this topic

- Until 2016, no studies have been done on simulating the respiratory motion via PKMs [1]
- After 2016, majority papers are for demonstrating chest compressions by PKMs [2]
- One such paper, has published study of the exact same task but in a scaled down version. [1]
- Regardless we have papers focusing on various aspects of the design such as -
    - Kinematics and statics [3]
    - Load capacity [4]
    - Control [5]
    - Overview [6]

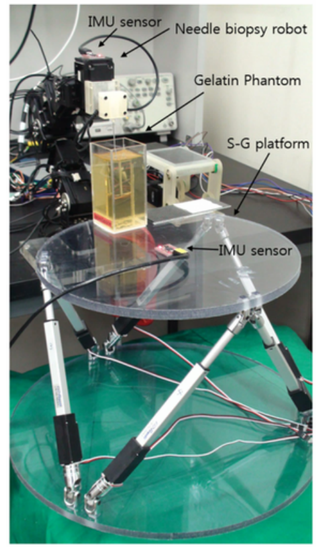
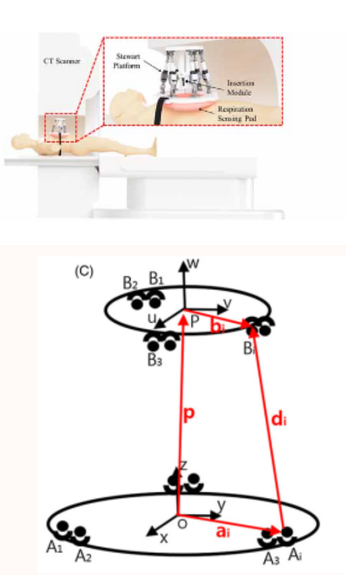


FIGURE 6 Test system (a) overview and (b) a gelatin phantom



REFERENCES

Impedance and admittance control for respiratory-motion compensation during robotic needle insertion – a preliminary test Yeoun Jae Kim | Jong Hyun Seo | Hong Rae Kim | Kwang Gi Kim * [1]

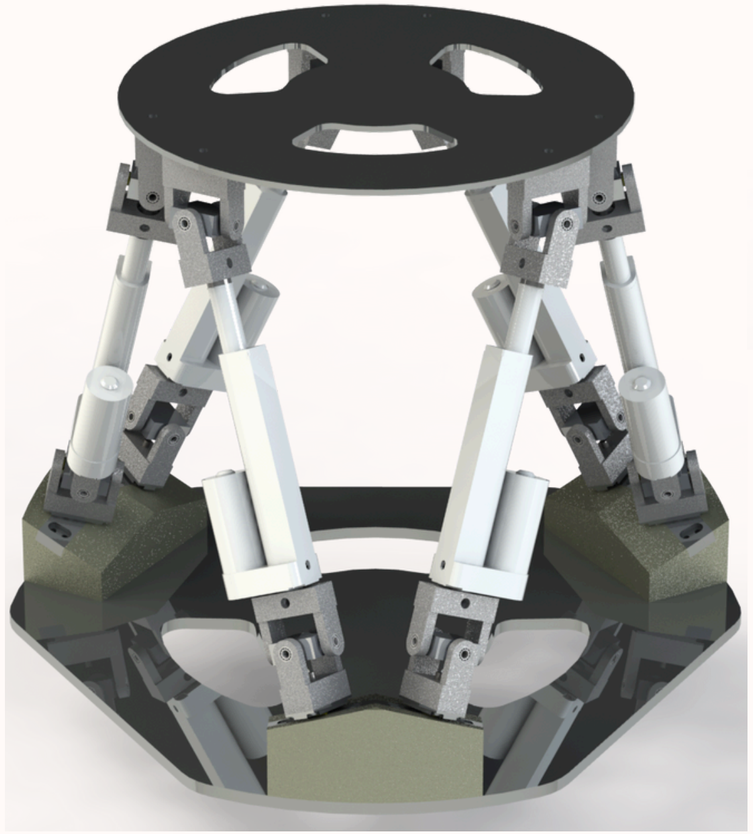
Autonomous Respiratory Motion Compensated Robot for CT-Guided Abdominal Radiofrequency Ablations : Anthony L. Gunderman; Mishek Musa; etc [2]

Kinematics and Statics of the Gough-Stewart Platform Jing-Shan Zhao * , Xiao-Cheng Sun and Song-Tao Wei [3]

Maximum Dynamic Load Carrying Capacity of a 6UPS-Stewart Platform Manipulator M.H. Korayem and M. Shokri1 [4]

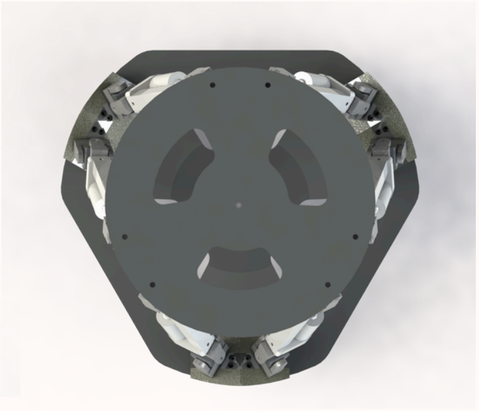
Stewart-Gough Platform: Design and Construction with a Digital PID Controller Implementation Flabio Dario Mirelez-Delgado, José Ronaldo etc [5]

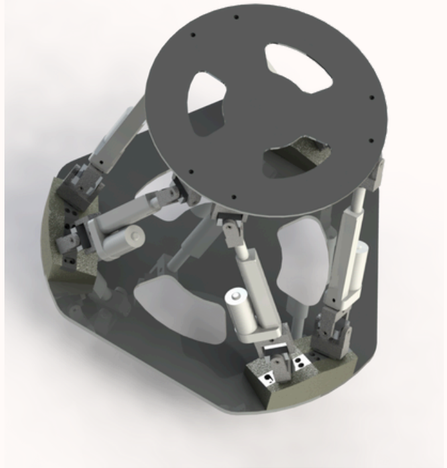
The Stewart platform manipulator: a review Bhaskar Dasgupta a,*, T.S. Mruthyunjaya b [6]

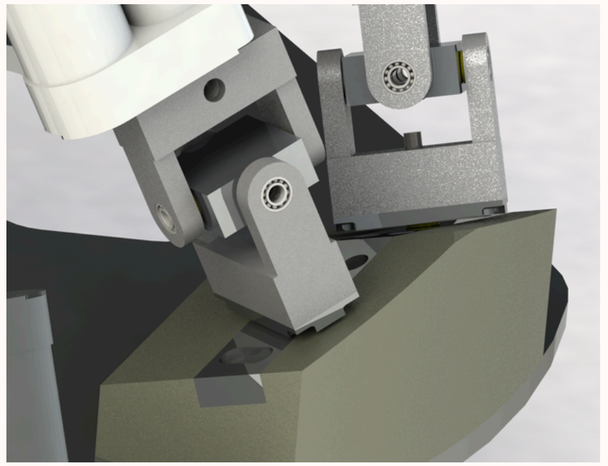


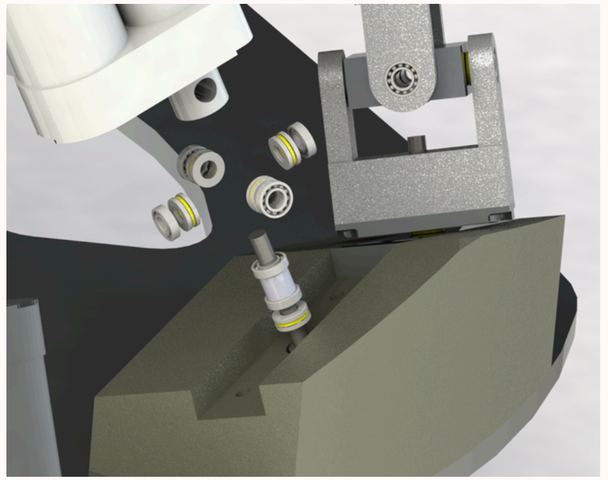




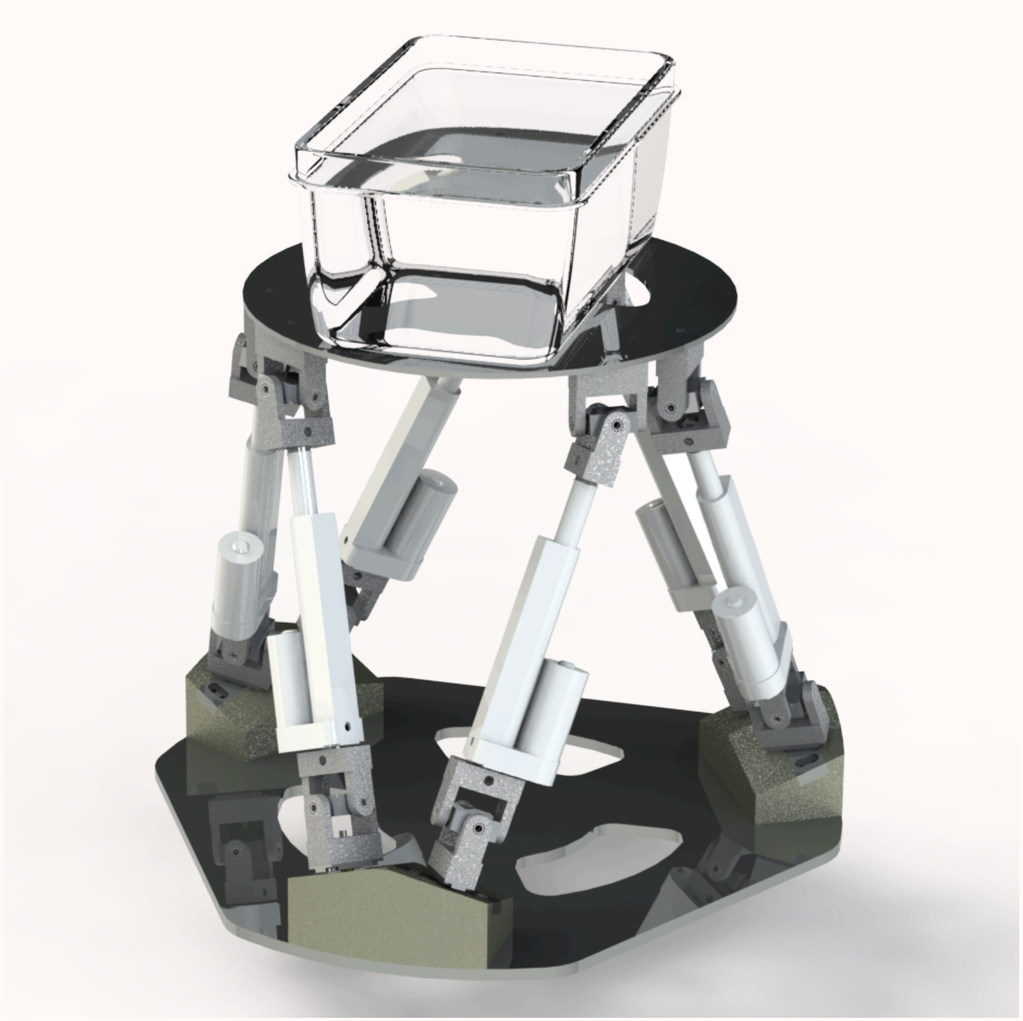


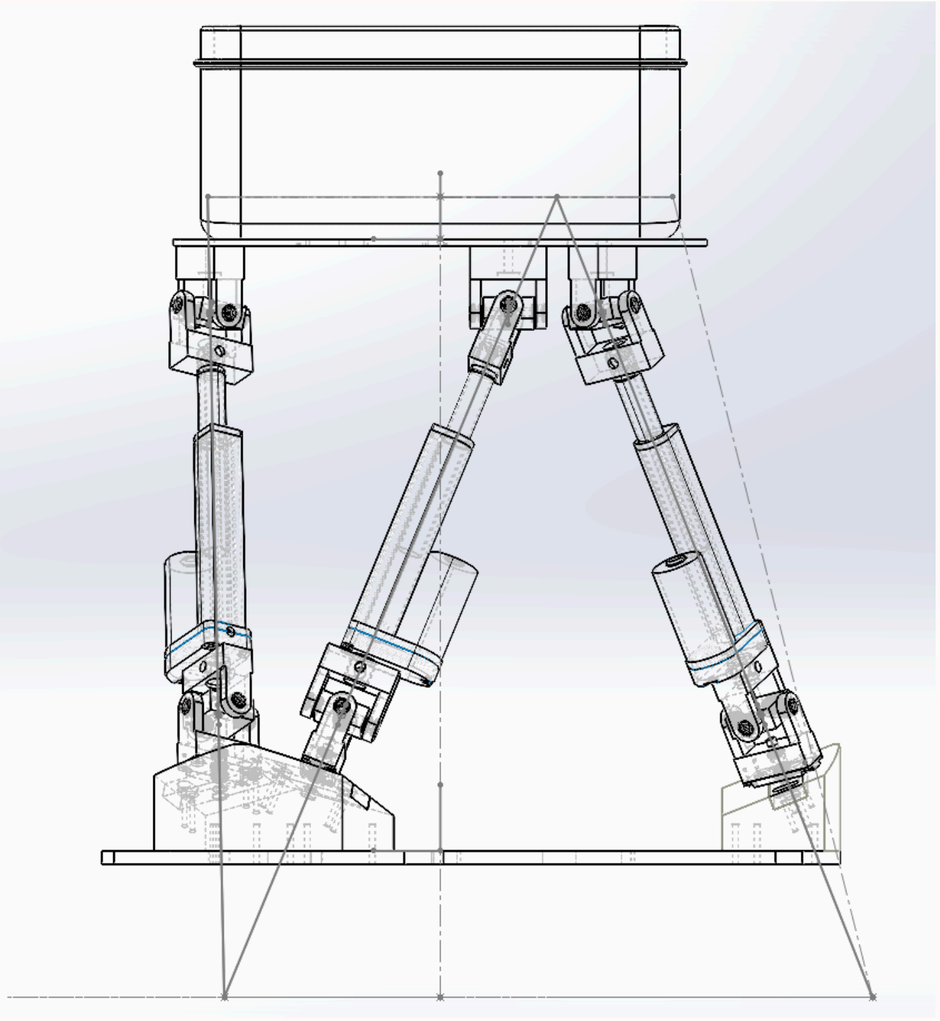






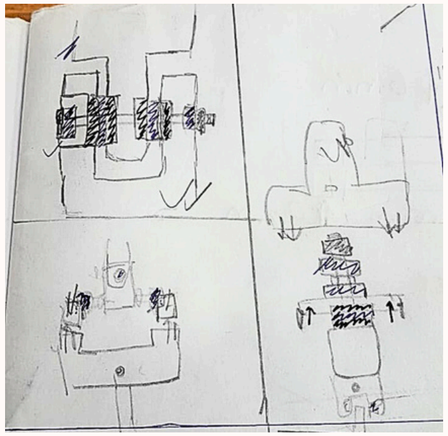
CAD MODEL

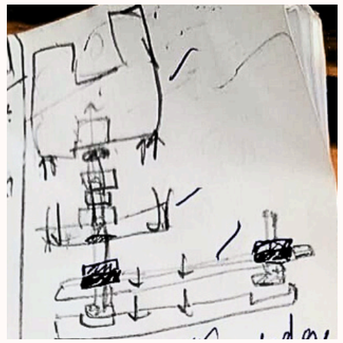


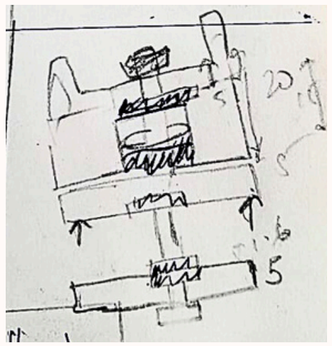


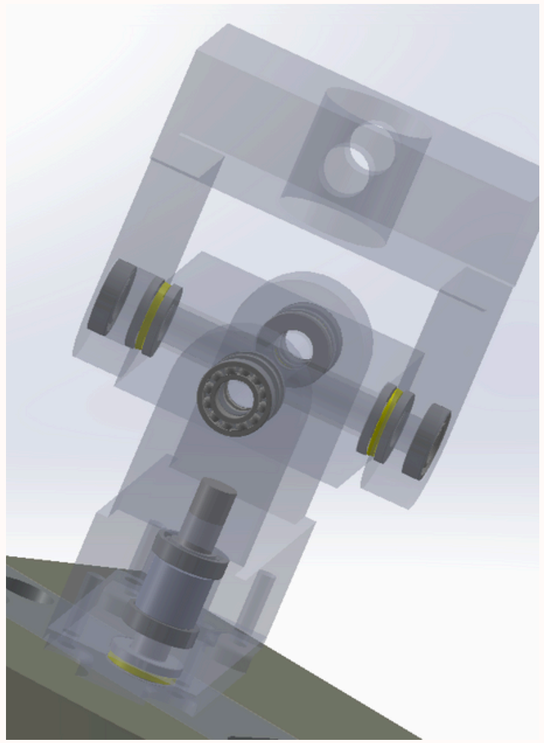
WITH PHANTOM

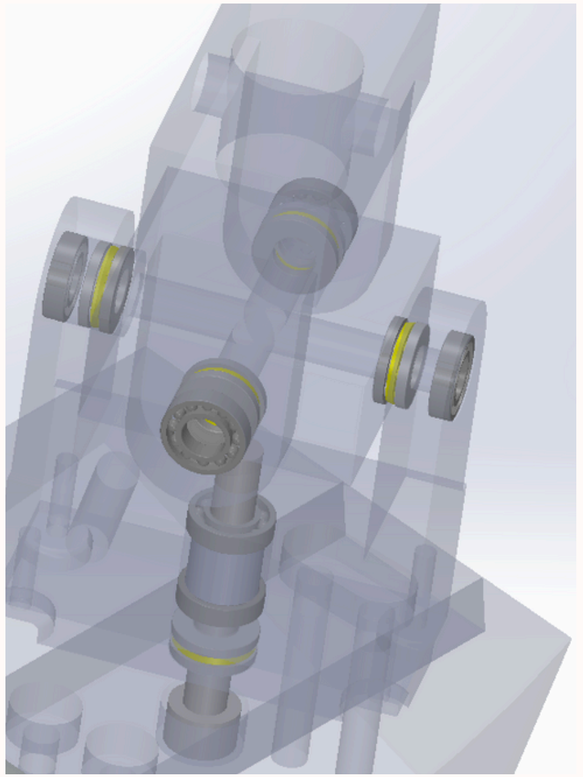
Self designed and assembled U-joints

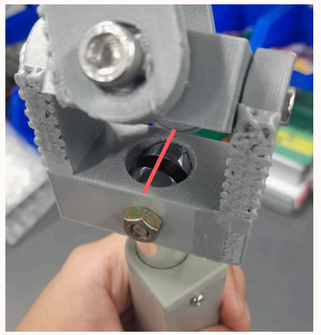




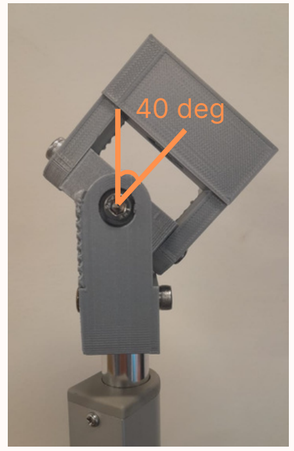








Joint might seem big but actually I have left little space only at the borders of the shaft. The shaft is thick

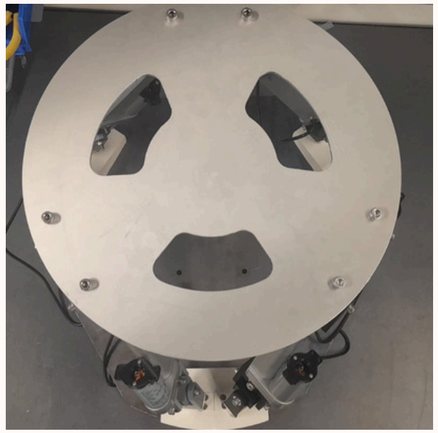


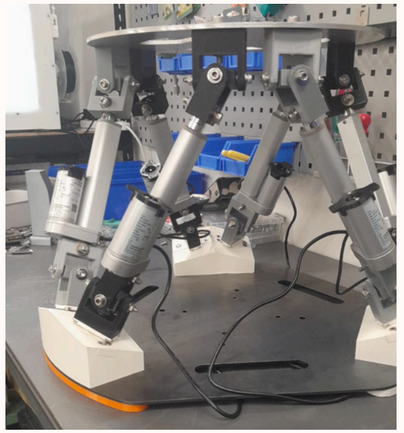
In total 108 bearings needed - 54 radial bearings(BR126) and 54 Thrust bearings (F6-12M)

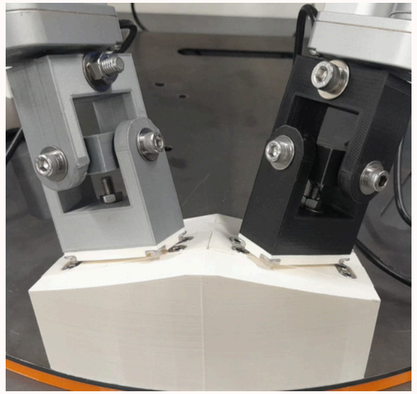
40 deg rotation is possible

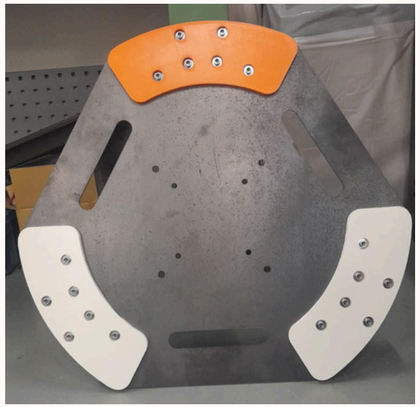


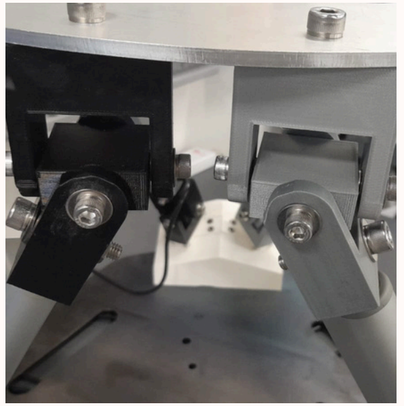












HARDWARE ASSEMBLED

Quantitatively

- Assembled **108** bearings
- Screwed **420** (258+162) times
- Lots of grinding (with machine)
- Lots of filing in and out
- Lots of gluing
- Assembled **341** (71+162+108) parts



PKM ROBOT FOR RESPIRATORY SIMULATION

BRIEF BACKGROUND

PKMs or parallel robots are the manipulators which having the end effector connected to the ground via several chains having actuations in parallel will have greater rigidity and superior positioning capability.. There are various configurations of this platform, but we would be looking at the 6 UPS (or 6 SPS) or commonly called the Stewart platform

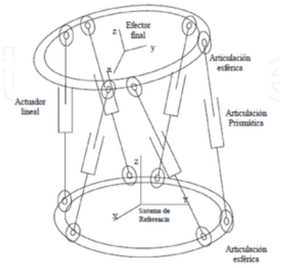
Our goal is to use this PKM to carry a phantom and simulate the breathing motion of a person. This would help us in robotic surgery research, as the new surgical tools to be tested will give be given a human centric phantom, in our case the abdomen surgery reaseach.

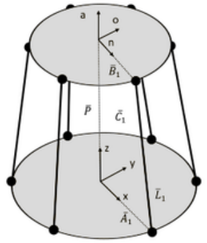


FINALISED CAD MODEL

IDEATION AND PROTOTYPES

Optimised the design for optimum range of motion, load capacity, size of bot enough for carrying a 20kg abdomen phantom. The design model contains over 108 bearings, 420 fastening joints and about 341 parts in total. The entire assembly was done under Prof. Ping Lang Yen's RMML lab at National Taiwan University.





P,θ,φ,ψ for any arbitrary position

$$\overline{L_i} = \overline{P} + \overline{B_i} - \overline{A_i}$$

$$\begin{bmatrix} L_{ix} \\ L_{iy} \\ L_{iz} \end{bmatrix} = \begin{bmatrix} P_{ix} \\ P_{iy} \\ P_{iz} \end{bmatrix} + \begin{bmatrix} B_{ix} \\ B_{iy} \\ B_{iz} \end{bmatrix} - \begin{bmatrix} A_{ix} \\ A_{iy} \\ A_{iz} \end{bmatrix}$$

$$\overline{A_i} = \begin{bmatrix} Acos(i_{th}) \\ Asin(i_{th}) \\ 0 \\ 1 \end{bmatrix} \qquad \overline{B_i} = \begin{bmatrix} Bcos(i_{th}) \\ Bsin(i_{th}) \\ 0 \\ 1 \end{bmatrix}$$

Home position

$$Rot(x,\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\theta & -sin\theta \\ 0 & sin\theta & cos\theta \end{bmatrix}$$

$$Rot(y,\phi) = \begin{bmatrix} cos\phi & 0 & sin\phi \\ 0 & 1 & 0 \\ -sin\phi & 0 & cos\phi \end{bmatrix}$$

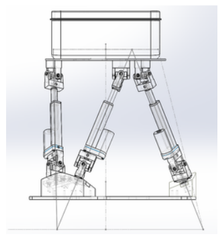
$$Rot(z,\psi) = \begin{bmatrix} cos\psi & -sin\psi & 0 \\ sin\psi & cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

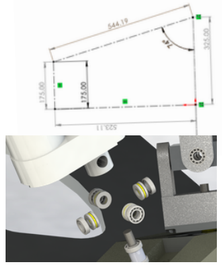
KINEMATICS OF 6 UPS PKM

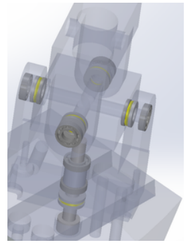
$$\begin{bmatrix} B_ix \\ B_iy \\ B_iz \\ 1 \end{bmatrix}_{rot} = [Rot(z,\psi)][Rot(y,\phi)][Rot(x,\theta)] \begin{bmatrix} B_ix \\ B_iy \\ B_iz \\ 1 \end{bmatrix}_{home}$$

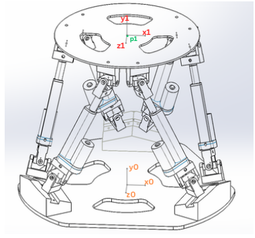
Rotation

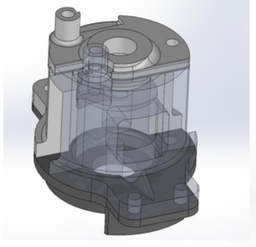
$$\begin{bmatrix} B_ix \\ B_iy \\ B_iz \\ 1 \end{bmatrix}_{rot} = \begin{bmatrix} cos\phi cos\psi & cos\psi sin\phi sin\theta - cos\theta sin\psi & sin\psi sin\theta + cos\psi cos\theta sin\phi & 0 \\ cos\phi sin\psi & cos\psi cos\theta + sin\phi sin\psi sin\theta & cos\theta sin\phi sin\psi - cos\psi sin\theta & 0 \\ -sin\phi & cos\phi sin\theta & cos\phi cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} B_ix \\ B_iy \\ B_iz \\ 1 \end{bmatrix}_{home}$$





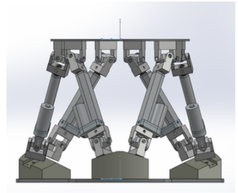


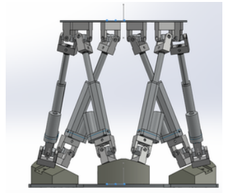


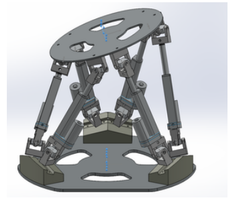


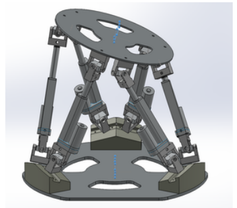


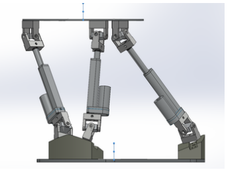
MODELLING AND OPTIMISATION OF PKM

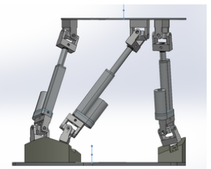




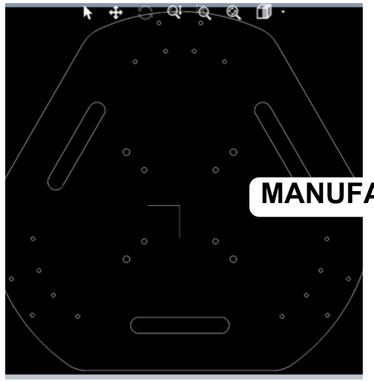




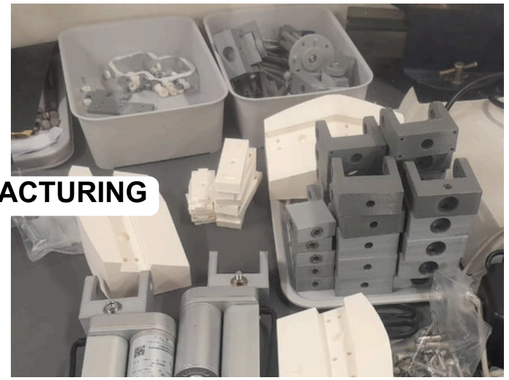




MANUFACTURING AND ASSEMBLY



MANUFACTURING

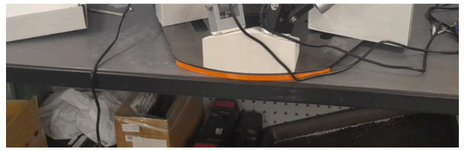




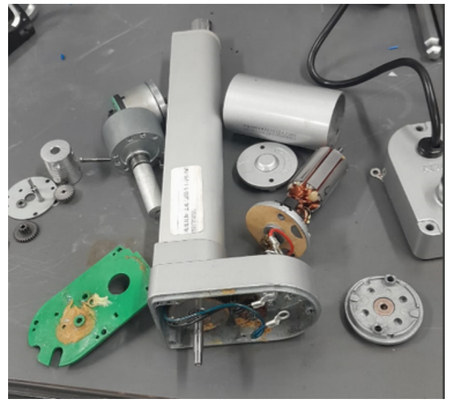


ASSEMBLY

108 BEARINGS, 420 FASTENING JOINTS AND 341 PARTS









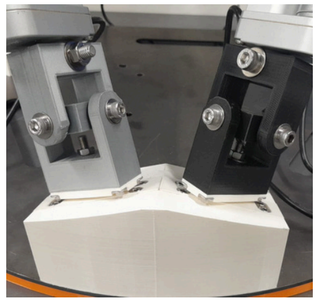


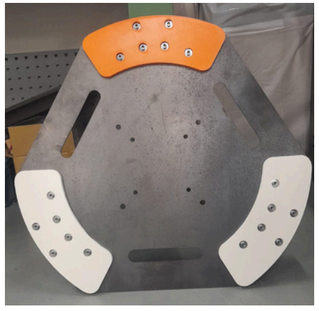


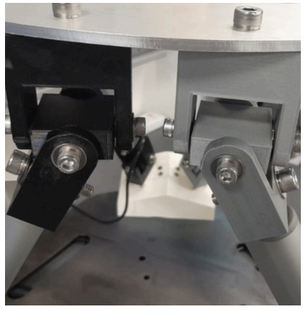


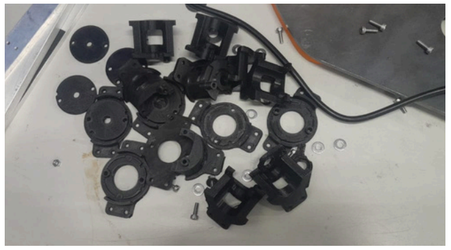










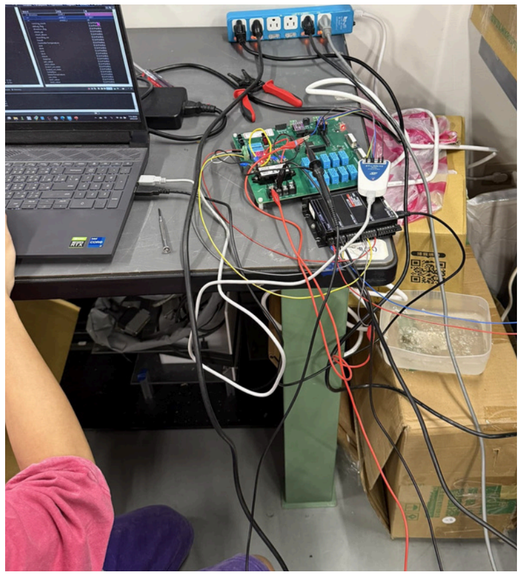


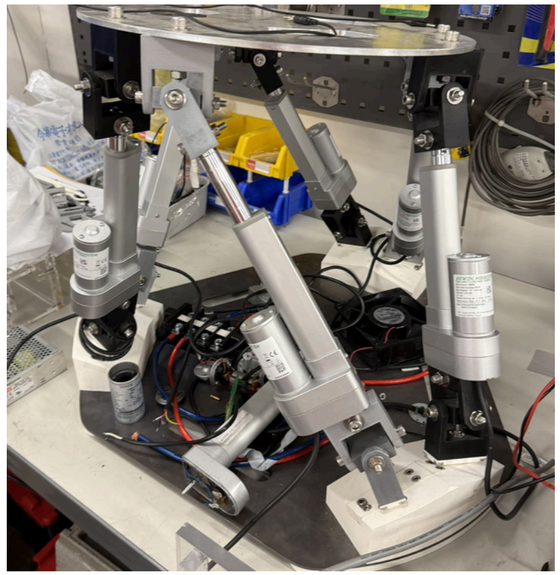
FINAL HARDWARE ASSEMBLY

FAILED PROTOYPES SENT FOR RE-DESIGN AND REWORK

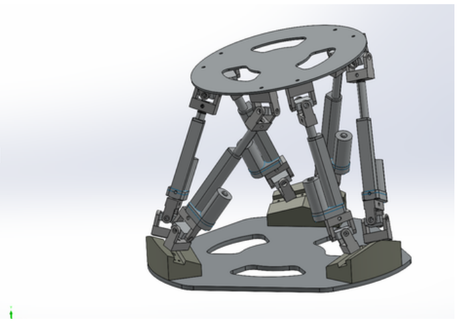
HARDWARE SIMULATION AND CONTROL

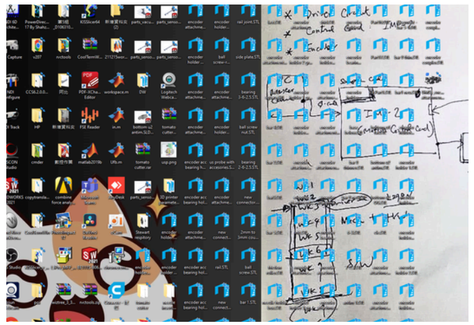
ELECTRONIC CIRCUIT AND CONTROL











The electronic circuit assembly was completed successfully, and seamless integration between the STM32 controller, driver circuitry, and actuators enabled stable and reliable platform operation.

COMPONENTS USED AND THIER PURPOSE

Component	Purpose
6 × HIWIN Linear Actuators	Provide linear motion for the six arms of the Stewart platform
STM32 MCU	Executes inverse kinematics and actuator control algorithms.
ST-LINK/V2	Programs and debugs the STM32 microcontroller.
Custom PCB	Interfaces the controller with drivers, sensors, and circuitry.
HIWIN Motor Driver Board	Drives the actuator motors from the STM32 MCU commands.
8-Channel Relay Section	Controls auxiliary loads and provides switching functionality.
Terminal Blocks	Provide secure electrical connections between components.
DC Power Supply Adapter	Supplies regulated DC power to the electronics and actuators.
Power Strip	Distributes AC power to the equipment on the workbench.
Oscilloscope Probe	Measures and monitors electrical signals for debugging.
Status LEDs	Provide visual indication of power and system states.
HIWIN Encoder	Provide position feedback for closed-loop actuator control.
Limit Switches	Establish reference positions and prevent overtravel.

CONCLUSION

The developed Stewart platform hardware and control system performed successfully, with the implemented control algorithms providing stable and accurate motion. Both the hardware setup and simulation model operated as intended, demonstrating the feasibility of the proposed design. The platform achieved the desired six degrees of freedom, and the obtained motion limits closely matched the design expectations.

The simulation results showed translational displacements of 403.55 mm to 514.31 mm along the Y-axis, −95.02 mm to 95.02 mm along the X-axis, and −101.63 mm to 101.63 mm along the Z-axis. Rotational capabilities of the platform were found to be −18.18° to 18.18° in roll, −17.54° to 17.54° in pitch, and −30.99° to 30.99° in yaw. These results confirm that the designed hardware and control algorithms were effective in achieving the required workspace and orientation ranges, validating the overall performance of the Stewart platform system.

THANK YOU!

